HARSHAL SURESH RAUT, Ph.D.

GPU / HPC COMPUTATIONAL SCIENTIST & ENGINEER

West Lafayette, IN | hraut@purdue.edu | LinkedIn | GitHub | Google Scholar | Portfolio

PROFESSIONAL SUMMARY

Computational scientist and engineer with 8+ years of experience developing high-performance numerical methods and physics-based simulation software in C/C++, CUDA, MPI, Fortran, and Python. Hands-on experience with GPU kernel optimization, CUDA-aware MPI, multi-GPU Krylov solvers, sparse linear algebra, OpenACC, PyTorch CUDA extensions, Nsight profiling, and production CFD/CFD-DEM solver development in MFiX-Exa/AMReX. Demonstrated ~85% parallel efficiency on 400 CPU cores, GPU acceleration on NVIDIA A100 systems, and 11 peer-reviewed journal publications.

TECHNICAL SKILLS

Programming: C/C++, CUDA C++, Python, MATLAB, Fortran
GPU & Parallel: CUDA, MPI/OpenMPI, CUDA-aware MPI, OpenMP, NCCL, OpenCL, Slurm, multi-GPU programming
Performance: Nsight Systems, Nsight Compute, GPU memory hierarchy, shared memory, streams, kernel profiling, benchmarking
Numerical / HPC: CG/PCG, BiCG/BiCGSTAB, Krylov methods, sparse CSR, cuSPARSE, PETSc, multigrid, finite difference/volume/element, projection methods
AI / Scientific ML: PyTorch/LibTorch, custom PyTorch CUDA extensions, TensorFlow, deep reinforcement learning, PINNs
Scientific Software: Linux, Git, CMake, MFiX-Exa, AMReX, OpenFOAM, CFDEM/LIGGGHTS, FEniCS/FEniCSx, ParaView, ANSYS Fluent/Icepak

PROFESSIONAL EXPERIENCE

Purdue University | West Lafayette, IN **Jan 2026–Present**
Postdoctoral Researcher — School of Mechanical Engineering

- Develop large-scale CFD-DEM models for particle-laden injectable formulations, coupling Eulerian fluid flow with Lagrangian particles, contact/cohesion models, and immersed-boundary treatment of a moving plunger.
- Develop finite-element pressure/transport solvers in FEniCS/FEniCSx for anisotropic biological tissue using MRI/DTI-derived tensor fields and coupled pressure/advection-diffusion physics.
- Implemented 90° rotational cyclic-periodic infrastructure in GPU-based MFiX-Exa/AMReX, including transformed scalar/vector mapping, particle wrapping, conservative seam transfer, and pressure-operator/multigrid treatment; validated quotient-operator action against full-cylinder references to 1.62×10^{-10}.

Johns Hopkins University | Baltimore, MD **Dec 2022–Dec 2025**
Postdoctoral Researcher — Flow Physics and Computation Laboratory

- Extended the VICAR3D immersed-boundary solver to variable-density free-surface flow with algebraic VOF, fifth-order WENO advection, density-weighted pressure projection, balanced-force surface tension, and moving-body/interface crossing.
- Parallelized production CFD/FSI workflows with MPI, achieving ~85% parallel efficiency on 400 CPU cores; executed large parameter studies with 450+ high-fidelity simulations.
- Developed force-partitioning and reduced-order analysis workflows linking coherent vortical structures to integrated hydrodynamic loads; created a public CPU/GPU rewrite (FlowFPM++) using MPI/OpenACC for NVIDIA GPUs.

Indian Institute of Technology Bombay | Mumbai, India **Jan 2018–Oct 2022**
Ph.D. Research Fellow — Computational Fluid Dynamics Laboratory

- Built in-house 3D multiphase/FSI solvers in C++/CUDA/MPI using sharp-interface level-set, immersed-boundary, pressure-projection, and phase-change algorithms.
- Developed GPU-accelerated pressure-Poisson/Krylov solver components using CSR sparse storage, warp-per-row SpMV, cuSPARSE triangular solves, and Nsight Systems/Compute profiling; tested on NVIDIA A100 GPUs.
- Developed a dual-grid DRL-CFD framework for active flow control and heat-transfer enhancement that reduced training cost by 5.5x relative to the fine-grid-only strategy.

SELECTED GPU / HPC PROJECTS

Multi-GPU Krylov Solvers for Implicit PDEs — GitHub | CUDA C++ | MPI | CUDA-aware MPI | A100

- Implemented BiCG, BiCGSTAB, and Jacobi solvers for a 2D transient heat equation with one MPI rank per GPU, direct GPU halo exchange, and CUDA kernels for domain updates.
- For a 1024×1024 grid over 1000 timesteps, repository benchmarks report BiCGSTAB at 0.58 s versus 6.05 s for Jacobi; includes Slurm job scripts for two A100 GPUs.

CUDA Scaled Dot-Product Attention + PyTorch Extension — GitHub | CUDA C++ | PyTorch | Shared Memory

- Implemented custom CUDA kernels for transformer attention and exposed them to Python through a compiled PyTorch CUDA extension.
- Used shared memory and custom kernel fusion/organization to reduce global-memory traffic; benchmarked the implementation against PyTorch scaled-dot-product attention.

Preconditioned Conjugate Gradient Solver in CUDA — GitHub | CUDA | cuSPARSE | CSR | Nsight

- Implemented sparse matrix-vector multiplication in CSR with a warp-per-row strategy and GPU triangular solves through cuSPARSE for a preconditioned Krylov solver.
- Profiled the solver with Nsight Systems/Compute and demonstrated >5x speedup on an NVIDIA A100 relative to the single-CPU-core baseline used in the original study.

c90-sector: 90° Rotational CFD-DEM Boundary Infrastructure — C++ | AMReX | GPU CFD-DEM | Multigrid

- Developed and verified rotational-sector coupling for a DOE exascale multiphase solver; public validation includes 1.62×10^{-10} quotient-operator agreement, particle/contact symmetry near roundoff, and full-cylinder-equivalent force-ledger closure.

Multiphase Free-Surface / Immersed-Boundary CFD — C++ | MPI | VOF | IBM

- Demonstrated a heaving/pitching foil repeatedly exiting and re-entering an air-water interface; showcase run maintains maximum post-projection divergence of 5×10^{-8} and liquid-volume change of 8.3×10^{-6}.

FlowFPM++ Force-Partitioning Solver — Fortran 2018 | MPI | OpenACC | NVIDIA A100

- Built a public CPU/GPU force-partitioning solver with PCG and OpenACC; at 512×512×64, A100 time-to-solution is 7.11 s versus 49.2 s on 16 CPU ranks, with CPU/GPU fields matching to roundoff.

EDUCATION

Indian Institute of Technology Bombay — Ph.D., Mechanical Engineering, 2022

Thesis: Computational Multi-Fluid Structure Interaction Study on Nucleate Boiling in Presence of Externally Actuated Surfaces | CGPA: 10/10

Indian Institute of Technology Hyderabad — B.Tech. (Honors) + M.Tech., Mechanical and Aerospace Engineering, 2016

Overall CGPA: 8.13/10

SELECTED PUBLICATIONS, PATENT & RECOGNITION

- Raut, H. S., Seo, J.-H., Mittal, R. “Hydrodynamics of flapping foils undergoing irregular motion with application to wave-assisted propulsion.” Journal of Fluid Mechanics, 2026.
- Raut, H. S., Bhattacharya, A., Sharma, A. “A dual grid-based deep reinforcement learning and computational fluid dynamics method for flow control and its application to nucleate pool boiling.” International Journal of Heat and Mass Transfer, 2024.
- 11 peer-reviewed journal articles; Indian Patent No. 507595 (issued 2024); Naik and Rastogi Award for Excellence in Ph.D. Research, IIT Bombay (2024).